Do Now · retrieval

Data types & unsigned binary

Name: ____________________ Class: _________ Date: __________

Answer four retrieval questions before reviewing earlier learning.

Q1

Choose the most suitable data type for 27 students and for a temperature of 18.5 °C .

Q2

A student ID is "0042" . Which data type would you use, and why would an integer be unsuitable?

Q3

Convert 00101101_2 into denary. Show the place values you add.

Q4

A student says: “The largest unsigned 8-bit value is 256 because $2^8 = 256$.” Explain the mistake.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Choose a type for the purpose — The symbols in a value do not decide its data type on their own. Think about what the value represents and what operations make sense.

Core idea 2

Binary → denary — Select only the place values whose bits are 1. — CODE:
00101101_2
$= 32 + 8 + 4 + 1$
$= 45$

Core idea 3

Denary → binary — Choose place values that add to the target. Write 1 in those columns and 0 in the others. — CODE:
$74 = 64 + 8 + 2$
$74_{10} = 01001010_2$

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Know the range — Eight bits give $2^8 = 256$ different patterns. The range begins at 0, so the largest unsigned value is $2^8 - 1$.

Vocabulary · use the correct technical terms

Integer — Which data type stores whole numbers, including zero and negative values?

Real / floating point — Which data type is suitable for numerical values that may contain a fractional part?

Character — Which data type stores one character such as a letter or symbol?

String — Which data type stores a sequence of characters such as an identifier?

Independent skill check · apply the lesson

Convert 37 to unsigned 8-bit binary; convert 01101001 to denary. Show the place values in both directions.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Binary → denary

Follow the worked method

1. Write the columns — CODE:
128 64 32 16 8 4 2 1
0 0 1 0 1 1 0 1

2. Use only the columns containing 1 — CODE:
32 + 8 + 4 + 1

3. Calculate the denary value — CODE:
32 + 8 + 4 + 1 = 45

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

YOUR TASK

Build the justification

CLUES · uncover one at a time

Clue 1: Is 0042 an amount to calculate with, or an identifier?

Clue 2: What would happen to its leading zeros if it were stored as an integer?

Clue 3: Choose a type that preserves the exact sequence of characters.

BUILD YOUR ANSWER · in BLACK PEN

Step 1: State the value's purpose.

Step 2: Choose the most suitable data type.

Step 3: Explain why a different type would change the identifier.

CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A weather-monitoring system
For each field, state a suitable data type and justify it:
number of readings collected;
air temperature, e.g. 18.5 °C;
a one-letter category, e.g. A;
station identifier "0042" ;
whether the sensor is working.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Choose a storage width
A system must store whole-number readings from 0 to 300 inclusive.
State the largest unsigned 8-bit value.
Explain why 8 bits are insufficient.
Determine the minimum number of bits required and justify it using the unsigned range.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The largest unsigned value that can be represented using 8 bits is 255.

Use key vocabulary from memory

1. Explain Integer in your own words.

2. Explain Real / floating point in your own words.

3. Explain Character in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

END SELF-CHECK · only after you have attempted the work

PAGE 5 · Guided We Do solution:

Use a string: "0042" is an identifier, and an integer would discard the leading zeros.

PAGE 3 · Independent skill check:

37 = 32 + 4 + 1, so 00100101. 01101001 = 64 + 32 + 8 + 1 = 105.

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-001/answers.html